

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1707

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution,

chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and

decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I J O Joanna Divine (192521003) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

J O Joanna Divine

Annexure A
CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

(a) Representation using Propositional Logic

Symbols

Let:

- F = Patient A has Fever
- C = Patient A has Cough
- B = Patient A has Breathlessness
- Flu = Patient A has Possible Flu
- M = Patient A has Possible Measles
- P = Patient A has Possible Pneumonia
- $\wedge$ = AND
- $\rightarrow$ = Implies / If-Then

Rules

1. Fever AND Cough $\rightarrow$ Possible Flu
2. Fever AND Rash $\rightarrow$ Possible Measles
3. Cough AND Breathlessness $\rightarrow$ Possible Pneumonia

Facts

(b) Forward Chaining

Forward Chaining starts with known facts and applies rules whose conditions are satisfied.

Step 1

Known facts:

Rule 1:

Since $F = \text{True}$ and $C = \text{True}$:

Step 2

Rule 3:

Since $C = \text{True}$ and $B = \text{True}$:

Step 3

Rule 2 requires:

But Rash is not given.

Therefore, Measles cannot be derived.

Final conclusions

(c) Backward Chaining for Pneumonia

Goal

Find a rule whose conclusion is Pneumonia:

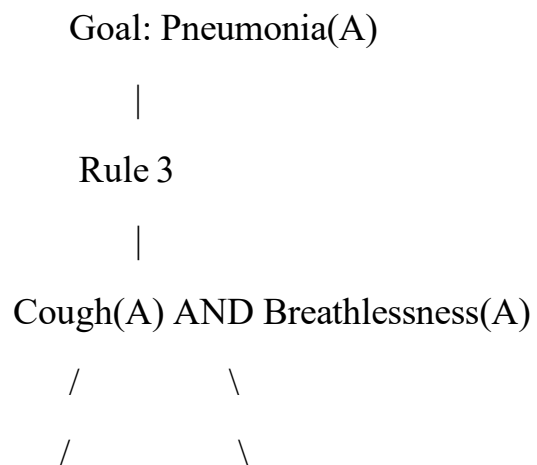
Therefore, reduce the goal into two sub-goals:

↓ Rule 3

Both are known facts:

Therefore:

Goal Reduction Tree



Cough(A) Breathlessness(A)

TRUE

TRUE

\

/

\

/

Goal Satisfied

|

Pneumonia(A)

Result

Patient A has Possible Pneumonia.

Case Study 2 — Autonomous Traffic Management Agent

Given FOL Knowledge Base

Rules

Rule 1

Rule 2

Rule 3

Facts

(a) Unification

For Rule 1:

Given facts:

Match:

Therefore:

Match:

with:

Again:

MGU

Applying the substitution:

is derived.

(b) Resolution Proof of Proceed(CarA)

Step 1: Convert relevant rules to CNF

Rule 1:

Using:

we get:

Rule 2

CNF:

Rule 3:

CNF:

Resolution Steps

From Rule 1 and the facts:

derive:

Using Rule 2:

Resolution gives:

Now Rule 3 becomes:

Given:

successive resolution gives:

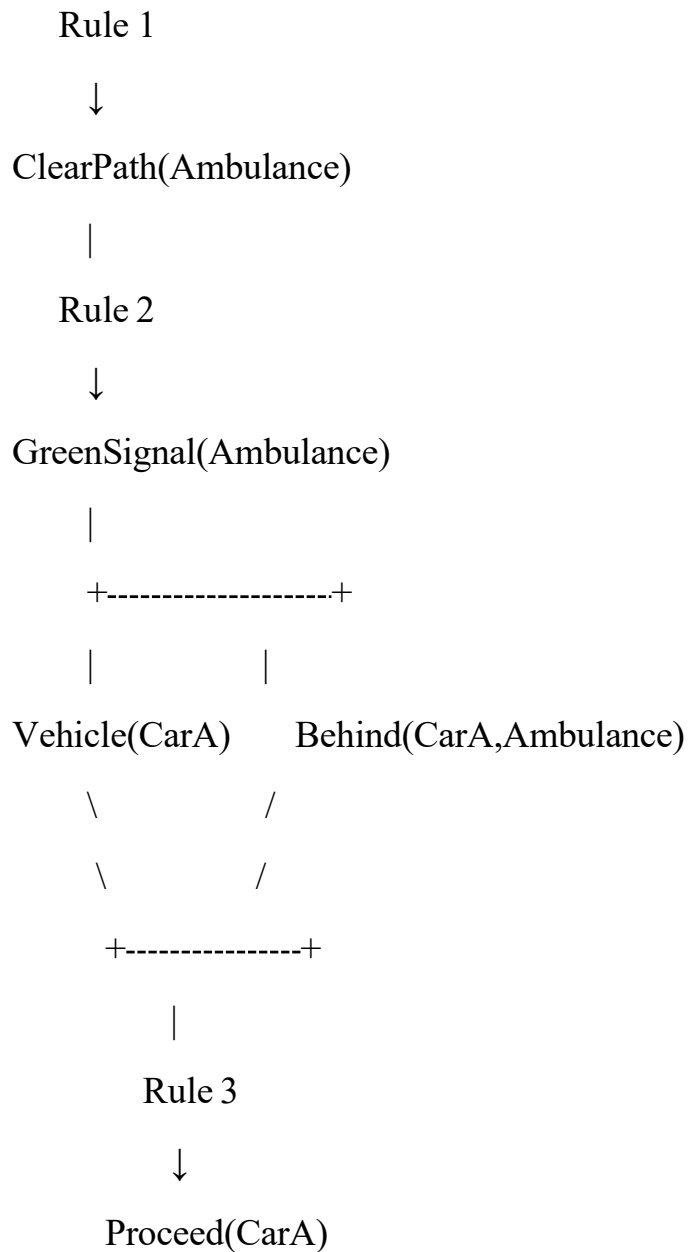
Resolution Chain

Vehicle(Ambulance)

+

EmergencyType(Ambulance)

|



Result

(c) Handling Two Simultaneous Emergency Vehicles

The current KB can identify an emergency vehicle individually because Rule 1 applies to every x:

Therefore, if we add:

the system can derive:

and then:

However, the KB does not specify what happens when two emergency vehicles require the same road or signal.

Additional rules needed

For example:

or:

Additional facts

For example:

Conclusion

The existing KB handles individual emergency vehicles but requires priority, route-conflict, and simultaneous-emergency rules to safely handle multiple emergency vehicles.

Case Study 3 — Agricultural Expert Reasoning System

Given KB

- 1.
- 2.
- 3.

Facts:

(a) CNF Conversion

P1

Original:

Eliminate implication:

Therefore:

P2

Original:

Eliminate implication:

Apply De Morgan's Law:

Therefore:

P3

Original:

Eliminate implication:

Using De Morgan's Law:

Therefore:

Facts in CNF

(b) Resolution Refutation for ApplyDripMethod

Goal

Negate the goal:

Add it to the KB.

Clauses

C1

C2

C3

C4

C5

C6

Resolution Step 1

Resolve C1 and C4:

Therefore:

Resolution Step 2

Resolve C2 with IrrigationNeeded:

Therefore:

Resolution Step 3

Resolve with C5:

Therefore:

Resolution Step 4

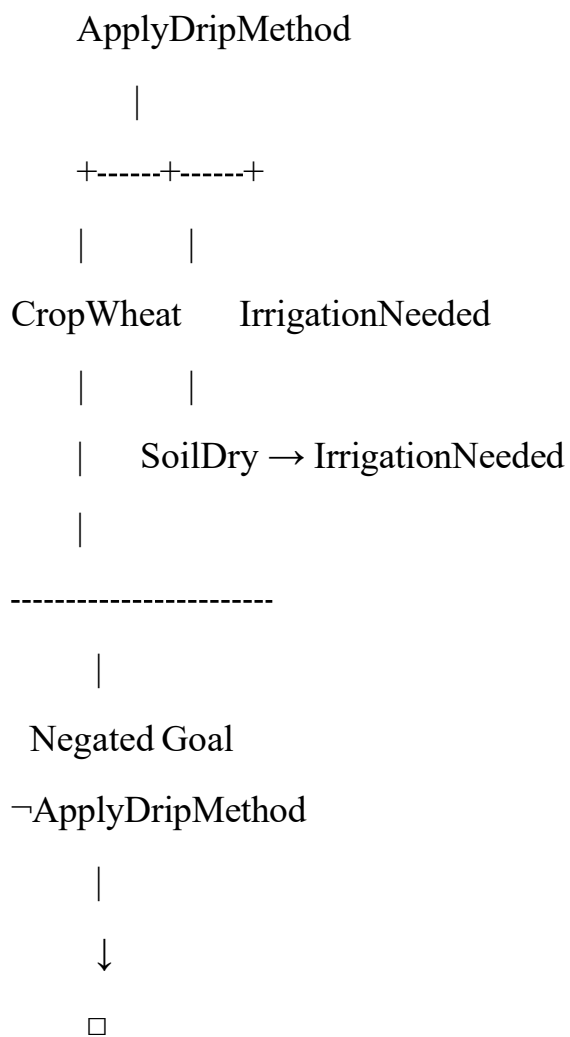
Resolve with negated goal C6:

Therefore:

The empty clause means contradiction.

Hence:

Resolution Proof Tree



Result

ApplyDripMethod is logically proven.

(c) Removing SoilDry

Suppose:

is removed.

Remaining fact:

Step 1

P1:

But SoilDry is no longer known.

Therefore:

cannot be derived.

Step 2

P2 requires:

Since IrrigationNeeded is unavailable:

cannot be derived.

Step 3

P3 requires:

We know CropWheat, but we do not know:

Therefore:

also cannot be logically derived.

Final conclusion

Without SoilDry:

- IrrigationNeeded $\rightarrow$ Not derivable
- ApplyDripMethod $\rightarrow$ Not derivable
- $\neg$ ApplyDripMethod $\rightarrow$ Not derivable

- CropAtRisk $\rightarrow$ Not derivable

Therefore:

The system needs an explicit fact/rule establishing that drip irrigation is not applied before concluding that the crop is at risk.

Case Study 4 — Wumpus World Knowledge Agent

Given Percepts

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

Rules

- 1.
 - 2.
 - 3.
 - 4.
-

(a) Belief State and Modus Ponens

Initial Facts

Step 1 — Stench

Rule:

Given:

Therefore:

Step 2 — Breeze

Rule:

Given:

Therefore:

Step 3 — Glitter

Rule:

For:

we get:

Therefore:

Derived Knowledge

Stench[1,2] → WumpusAdjacent[1,2]

Breeze[1,1] → PitAdjacent[1,1]

Glitter[2,2] → Gold[2,2]

(b) Forward Chaining for Possible Wumpus and Pit Locations

Wumpus

From:

we know:

The adjacent cells of [1,2] are:

Therefore possible Wumpus locations are:

Pit

From:

we know:

The adjacent cells are:

Therefore possible Pit locations are:

Forward-Chaining Summary

Stench[1,2]



Wumpus adjacent to [1,2]



Possible Wumpus:

[1,1], [1,3], [2,2]

Breeze[1,1]



Pit adjacent to [1,1]



Possible Pit:

[1,2], [2,1]

Glitter[2,2]



Gold[2,2]

(c) Safety of Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$

The proposed path is:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

Cell [1,1]

The KB tells us that there is a Breeze at [1,1].

Therefore, a pit may be in an adjacent cell.

However, there is no direct evidence that [1,1] itself contains a pit.

So:

Cell [1,2]

There is a Stench at [1,2].

Therefore, a Wumpus is possible in one of its adjacent cells:

The KB does not establish that [1,2] itself contains a Wumpus.

However, [1,2] is a possible pit location based on the Breeze at [1,1].

Therefore, [1,2] is not proven safe.

Cell [2,2]

We know:

But [2,2] is also a possible Wumpus location because it is adjacent to [1,2].

Therefore, [2,2] is not proven safe.

Final Decision

The rule for safety requires:

But the KB does not provide these negative facts for the cells in the proposed path.

Therefore:

The agent should not assume the path is safe based only on the given percepts.

Overall Conclusion

These four case studies demonstrate how logical agents use different reasoning techniques:

Case Study	Main Technique	Result
Smart Medical Diagnosis	Forward & Backward Chaining	Pneumonia derived
Traffic Management	Unification & Resolution	Proceed(CarA) proved

Case Study	Main Technique	Result
Agricultural System	CNF & Resolution Refutation	ApplyDripMethod proved
Wumpus World	Modus Ponens & Forward Chaining	Gold identified, but path not proven safe

Conclusion: Logical agents can derive new knowledge from existing facts and rules using techniques such as Forward Chaining, Backward Chaining, Unification, CNF conversion, Modus Ponens, and Resolution. These methods provide a systematic approach for decision-making in medical, traffic, agricultural, and game environments.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand